Chapter 1 Laplacian

List of Notions:

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- Local integrable functions

$$L^1_{loc}(\Omega)$$

1.1 The Laplace Operator

Definition 1.1.0.1 (The Laplace Operator) In $\mathbb{R}^d$, define the Cartesian coordinates $x = (x^1, \ldots, x^d)$. Now, let

$$\Delta f = \sum_j \frac{\partial^2 f}{\partial x_j^2}$$

$f = f(x_1, \ldots, x_d)$ is twice differentiable function. This operator is called **Laplace operator**.

Some of equations:

- Wave equation

$$\frac{\partial^2 U}{\partial t^2} = \Delta U(t, x)$$

  Here, $U(t, x)$ denotes the displacement from the equilibrium of the vibrating object at the point $x \in R^d$ at time t.

- Heat equation

$$\frac{\partial U(t, x)}{\partial t} = \Delta U(t, x)$$

- Laplace equation

$$\Delta U(t) = 0$$

- Poisson equation

$$-\Delta U(x) = f(x)$$

- Schrödinger Equation

$$i \frac{\partial U(t, x)}{\partial t} = -\Delta U(t, x)$$

1.1.1 The Laplcian on a Riemannian Manifold

1.2 The Spectral Theorems

1.2.1 Weak Derivatives, Sobolev Spaces

Weak Derivatives

Definition 1.2.1.1 Let $\Omega \subseteq \mathbb{R}^d$ be a domain, $u, v \in L^1_{loc}(\Omega)$, suppose that for any $\varphi \in C^1_0(\Omega)$

$$\int_\Omega u\partial_j \varphi dx = -\int_\Omega v\varphi dx$$

where $\partial_j = \frac{\partial}{\partial x_j}$.

We say: $\partial_j u$ exists in Ω in the **weak sense**;equals to v.

Remark Suppose that $u \in C^1(\Omega)$. With the integration by part, we have

$$u\partial_j\varphi + \partial_j u\varphi = \partial_j(u\varphi)$$

Then,

$$\int_\Omega u\partial_j\varphi dx = \int_\Omega \partial_j(u\varphi)dx - \int_\Omega \varphi\partial_j u dx$$

And with Gauss's theorem, we have

$$\int_\Omega \partial_j(u\varphi) = \int_{\partial\Omega}(u\varphi)\cdot v_j dS$$

Suppose that φ is a test function, then,

$$\varphi|\partial\Omega = 0 \Rightarrow (u\varphi)|\partial\Omega = 0$$

Thus, we can assumpt, that

♠ **Proposition 1.2.1.2** For $\Omega \subseteq \mathbb{R}^d$ we have

Sobolev Space

Definition 1.2.1.3 Set $H^0(\Omega) = L^2(\Omega), m \in N$. The **Soblev space** $H^m(\Omega)$ is defined as following:

$$H^m(\Omega) := \{u \in L^2(\Omega)|\partial_j u \text{ exists in the weak sense}, \partial_j u \in H^{m-1}(\Omega), j = 1, \dots, d\}$$

In general, if we take the following notion

$$\int_\Omega u(D^\alpha\varphi)dx = (-1)^{|\alpha|}\int_\Omega v\varphi dx, \varphi \in$$

Then, v is called the **weak derivative** of order α. Denoted by $D^\alpha u = v$.

$$W^{k,p}(\Omega) := \{u \in L^p(\Omega)|D^\alpha u \in L^p(\Omega)\}$$

- $H^m(\Omega)$ is equipped with the inner product

$$\begin{aligned}(u,v)_{H^1(\Omega)} &:= \int_\Omega uv dx + \int_\Omega < \nabla u, \nabla v > dx \\ &= \int_\Omega uv dx + \int_\Omega \sum_j \frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_j} dx \\ &= \int_\Omega uv dx + \sum_j \int_\Omega \frac{\partial u}{\partial x_j}\frac{\partial v}{\partial x_j} dx\end{aligned}$$

Then,

$$(u,v)_{H^m(\Omega)} = (u,v)_{L^2\Omega} + \sum_j \int_\Omega (\partial_j u, \partial_j v)_{H^{m-1}(\Omega)}, m \geq 2$$

The induced norm

$$\begin{cases} \|u\|^2_{H^1(\Omega)} = \|u\|^2_{L^2(\Omega)} + \|\nabla u\|^2_{L^2(\Omega)} \\ \|u\|^2_{H^m(\Omega)} := \|u\|^2_{L^2(\Omega)} + \sum_j \|\partial_j u\|^2_{H^{m-1}(\Omega)}, m \geq 2 \end{cases}$$

Thus, the Sobolev space is a Hilbert space.

Weak Solutions

◇ **Lemma 1.2.1.4** ...

1.2.2 The Weak Spectral Theorem for Dirichlet Laplacian

1.2.3 Self-Adjoint Unbounded Linear Operators

1.2.4 The Dirichlet Laplacian via the Friedrichs Extension

1.3 Elliptic Regularity, and Strong Spectral Theorems

Chapter 2 Variational Principles and Applications

2.1 Variational Principles for Laplace Eigenvalues

2.1.1 The Rayleigh Quotient

Definition 2.1.1.1 (Semi-Bounded Quadratic Form) Let $\mathcal{H}$ be a real Hilbert space, with inner product $(-,-)_{\mathcal{H}}$. Consider the symmetric bilinear form

$$\mathcal{L} : U \times U \to \mathbb{R}; U := Dom(\mathcal{L}) \qquad \text{dense in } \mathcal{H}$$

We say that **a quadratic semibounded from below** if there exists a constant $c \in \mathbb{R}$, such that

$$\mathcal{L}[u,u] \geq c(u,u)_{\mathcal{H}}, u \in Dom(\mathcal{L})$$

2.2 Consequences of Variational Principles

2.3 Weyl's Law and Polya's Conjecture

2.4 Eigenvalue Inequalities

Chapter 3 The Steklov Problem, and the Dirichlet-to-Neumann Map